



such as the grass snake, specialize in catching aquatic animals. The anaconda, the heaviest snake in the world, uses water for cover and for support: despite its size, water buoys it up, reducing its effective weight to almost nothing and enabling it to swim at considerable speed.

In addition to providing food, freshwater is a valuable resource for some land-based animals seeking refuge from predators or



ROOSTING AMONG THE REEDS

Swallows, starlings, and other flock-forming songbirds often use reeds as overnight roosts. The sleeping birds are relatively safe from predators because reeds are usually surrounded by water.

daytime heat. At dusk, birds often roost on lakes and reservoirs while, during the day, hippos, capybaras, and beavers spend a lot of time in the water, emerging under the cover of darkness to feed on land.

Surviving drought

In warm parts of the world—particularly in the tropics—rivers and lakes often dry out completely for several months each year. In these conditions, freshwater animals need some unusual adaptations to survive.

When a lake's water level drops, its temperature increases. Warm, stagnant water often contains so little oxygen that many fish suffocate, but lungfish—characteristic inhabitants of tropical lakes and wetlands—are experts at dealing with drought. They gulp air at the surface and, when their home starts to dry out, they burrow into the mud, sealing themselves inside mucous cocoons. Later in the year, when heavy rain falls and soaks the mud, the cocoon breaks down, and the fish wiggle away.

DRY TIMES

Two staring yellow eyes identify a common caiman in the mud of a dried-out lake. During drought, caimans can survive being entombed for many months as long as the mud stays moist.

Caimans and turtles also hide away in this way, although their scaly, waterproof skins mean that cocoons are not needed. Being “cold-blooded,” or ectothermic, they need relatively little energy to stay alive, so they can survive drought-induced food shortages for months.

Animals that cannot survive drought often leave behind drought-resistant eggs, which hatch when water returns—an effective way of bridging the gap between one wet period and the next. This survival strategy is used by water fleas, rotifers, tardigrades, and many other “micro-animals,” as well as some kinds of fish, which live in temporary pools or in the film of freshwater that covers mosses and other plants.



BREATHING AT THE SURFACE

Mosquito larvae hang upside down at the water's surface, breathing through snorkel-like tubes. The tubes are tipped with water-repellent hairs that break through the surface film.

Coming up for air

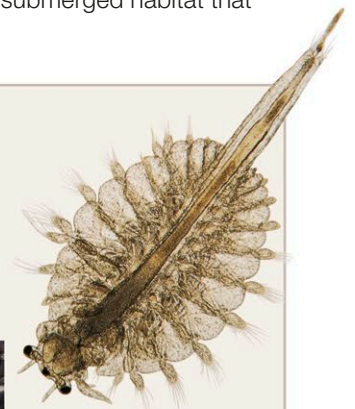
Freshwater animals need to breathe oxygen. Fish use gills to extract it from the water, but many invertebrates collect it from the air.

These air breathers include many water snails, many insects, and the water spider—the only spider to have evolved a fully aquatic lifestyle.

For animals living at the water's surface, air is easy to reach. For fully submerged ones, breathing requires periodic trips to the surface to replenish air reserves, which are stored in or on the insect's body, often forming a film-like bubble that gives the animal a silvery sheen. The water spider has an exceptionally elaborate storage system. It constructs a “diving bell” from strands of silk, trapping a large bubble of air inside it. The bell acts as both lair and nursery—a unique example of an animal creating a submerged habitat that resembles dry land.

LIVING IN SALT LAKES

Salt and soda lakes are the most saline habitats on earth, with up to 10 times as much dissolved salt as the sea. They form where rainfall is low and evaporation rates are high because temperatures often climb to over 104°F (40°C). Relatively few animals can deal with these extreme conditions, but the ones that do can be extremely numerous because they face very little competition. Salt-lake food chains are based on cyanobacteria—plantlike microorganisms that harness the energy in sunlight, turning it into food.



LIVING IN BRINE

Brine shrimps are small crustaceans that thrive in salt-rich desert lakes. They breed rapidly, producing eggs that can withstand several years of drought.

FILTER FEEDERS

Lesser flamingos use their beaks to filter microorganisms out of the water. A large colony can consume more than one ton of cyanobacteria a day.